Economic Sustainability and Social Equality in the Technological Era – the Authors
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The effect of quality of information technology disclosure and open innovation on firm value companies in the financial sector companies in Bei period 2020–2022

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ABSTRACT: This study aims to test and obtain empirical evidence that the Quality of Information Technology Disclosure and Open Innovation has an effect on Firm Value. The objects of this research are all financial sector companies listed on the Indonesia Stock Exchange in 2020–2022. The sampling technique used is a purposive sampling method with a total sample of 36 companies. The data analysis technique in this study used the panel data regression testing method with the STATA statistical tool version 15. The results of this study partially show that the quality of information technology disclosure has no effect on firm value, and open innovation has an effect on firm value. The results of the study simultaneously show that together the variables quality of information technology disclosure and open innovation have a significant effect on firm value.

Keywords: Information Technology Disclosure Quality, Firm Value, Open Innovation

1 INTRODUCTION

The development of information technology and innovation has changed people's lifestyles and the world of work. The urge to create innovations that provide *convenience* for people has also affected the financial sector. Apart from providing online services and launching mobile apps, digital transformation in the financial sector also focuses on improving customer experience by integrating digital technology into interactions with them (Mutiasari 2020).

According to MNC Sekuritas analyst, Tirta Widi Gilang Citradi in the online news page Liputan6.com, in 2020–2021, stocks related to Old Money were not in high demand because these companies were suffering from the pandemic. So investors started to target New Economy stocks such as digital banks whose share prices had boomed even when the digital banks didn't have digital application products yet. In 2022, the factors that supported the rise of digital bank stocks began to fade and the products or technology supplied were unable to keep up with the increasing demand trend. As a digital bank, such as PT Bank Jago Tbk (ARTO) with a stock price decreased from Rp16,000 to Rp3,720, PT Bank Raya Tbk (AGRO) with a closing stock price from Rp1,810 to Rp404 and PT Bank Neo Commerce Tbk (BBYB) with a closing stock price from Rp2,630 to Rp645. The phenomenon of digital trends that have become stronger since the pandemic has had an impact on investor interest and the share prices of companies in the financial sector.

Since 2019, banks have been adapting and starting to use the Open Banking system, which is a system that provides data on financial institutions and can be accessed by various parties such as application developers and fintech through the Application Programming Interface

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(API) (Muqorobin *et al*. 2021). The system allows banks to collaborate and partner with businesses such as fintech in developing new financial services. As an example of collaboration, PT Bank Rakyat Indonesia Tbk in collaboration with PT Visionet Internasional (OVO) launched the OVO U Card credit card for millennials. In addition, PT Bank MNC Internasional Tbk partnered with Kredit Pintar to provide lending features in the MotionBanking application. With this collaboration phenomenon, banks are not only relying on their internal resources but also working with external partners to improve the capabilities and quality of financial services offered to the public.

Information technology investment is the expenditure of a large number of funds to add or invest in IT including products and services such as hardware, software, telecommunications, human resource development and training, IT consulting and outsourcing. (Farouk and Dandago 1970). The use of IT can change the way companies compete in maintaining their business (Alalie *et al*. 2018). Companies that follow technological developments will adopt IT as an opportunity to gain profits with the products or services provided.

Open Innovation is a process where companies start using external knowledge to facilitate innovation activities and establish relationships and cooperation with external partners (Sengupta and Sena 2020). Many firms are now switching from closed innovation to open innovation, where firms start looking for entities as business partners in achieving profits in the development of old product innovations and the creation of new products. In addition, intangible assets play an important role in the success of open innovation both as a product and as a facilitator of innovation because the actors involved in open innovation are influenced by intangible features such as human resources (consultants) and partnerships in collaboration activities. (Grimaldi *et al*. 2017).

If the investment is made by a company in the field of information technology to advance the company, the investment may used to ensure the stability and sustainability of a business (Karim *et al*. 2022). The company's annual report that discloses information technology can be a signal for external parties. The signal reflects the company's ability to use technology to improve service quality to customers and maintain data security. This is a consideration for external parties in making investments.

Access to external resources and collaborating with external partners is an important element as it allows for a combination of new resources and more agile commercialization of innovations (Witell *et al*. 2017). Open innovation as a positive signal shows openness to ideas, collaboration ability, and the company's commitment to innovation. With open innovation, companies can develop old products and create new products through collaboration to increase firm value.

There is previous research that results in information technology affecting firm value such as research by Risa and Hermanto (2023) dan Bardhan *et al*. (2013). In contrast to the research conducted by Andrayani and Dewi (2014) which found that IT Spending does not affect firm value.

Research conducted by Emeneka and Okerekeoti (2022) dan Berzkalne and Zelgalve (2013) found that innovation in intangible assets affects firm value. In contrast to research conducted by Fitri *et al*. (2023) dan Ionita and Dinu (2021) which found that innovation does not affect firm value.

This study aims to reveal how much the role of the quality of Information Technology disclosure and Open Innovation on Company Value, from the description above there are several differences in the results of previous studies, so it encourages researchers to research to strengthen previous research using financial sector companies listed on the IDX for the period 2020–2022.

1.1 *Theoretical studies signaling theory*

This theory describes the behavior of two parties when there is a difference in access to information, with signaling to refer to actions taken by the signal owner to influence the views and behavior of the signal receiver (Spence 1973). A signal can be an observable

action that aims to show the hidden characteristics of the signaler and the type of information that is considered a positive signal for external parties is the annual report or financial report (Ulum *et al.* 2021). Based on the explanation above, it can be concluded that signaling theory is a theory that explains that the sender sends signals in the form of information that describes the current condition and sustainability of the company in the future.

Theoritical studies signaling theory

1.2 *Quality of information technology disclosure*

Information technology is a technological tool used by a company to create, process and transmit information in various forms (Primawanti and Ali 2021). Jadi, Information technology is considered important as a tool that supports companies in providing services to customers and accelerating the transformation towards digital companies that will attract public attention in purchasing stocks and increasing firm value.

1.3 *Open innovation*

According to Chesbrough (2003), open innovation is defined as a process in which a company can get ideas, technology, knowledge, and skills from sources external to the company. External sources can share new ideas with internal sources, and vice versa. By collaborating with external partners, companies can utilize the partner's knowledge to create new innovative products or develop new services that attract investors to invest. Andrews and De Serres (2012) state that innovation-based growth is supported by intangible assets including organizational knowledge and various kinds of intellectual property. In addition, intangible assets such as computer software, patents, copyrights, customer lists, licenses, customer relationships and supplier relationships are considered indicators of innovation as well (Berzkalne and Zelgalve 2013).

1.4 *Firm value*

Firm value is the expectation of investors as the party who invests in the company, usually, investors see the value of the company from the stock price (Maharani and Puspitasari 2022). The better the management's ability, the better the company will signal to investors and stakeholders. It is expected that the market will react and influence the stock price.

2 FORMULATION OF HYPOTHESIS

2.1 *Quality of information technology disclosure on firm value*

Every company is required to be able to provide solutions and benefits of the products and services to the public as a form of their seriousness towards the sustainability of their business (Pertiwi and Suhartini 2022). The disclosure of the company's information technology investment reflects a positive signal that the company is serious about developing its business in the era of digital technology that affects society's daily activities.

Furthermore, based on research conducted by Risa and Hermanto (2023) states that the better and more advanced the technology used by the company, the more people are interested in the company. This means that if the company does not keep up with the development of information technology, the company will lose interest from the public which can affect the firm's value. Previous research has found that information technology affects firm value (Bardhan *et al.* 2013; Risa and Hermanto 2023).

H_1: The Quality of Information Technology Disclosure affects Firm Value.

2.2 *Open innovation on firm value*

The company will receive positive feedback from investors if the innovative actions taken have an impact on the company's profitability so the feedback also increases the firm's value (Fitri *et al*. 2023). Innovative actions are carried out to assist companies in fulfilling customer needs to gain investor trust and interest in the company.

Collaboration with external partners such as suppliers, customers, and other companies tends to be beneficial for companies, especially in terms of innovation due to a combination of complementary capabilities and a common goal of developing new services (Mazzola *et al*. 2016). The collaboration that companies do is a sign to external parties regarding the company's ability to fulfill customer needs for digital services. This is expected to attract many communities to invest and has a positive impact on firm value. The results of previous studies found that innovation affects firm value (Berzkalne and Zelgalve 2013; Choi *et al*. 2022; Emeneka and Okerekeoti 2022).

H_2: Open Innovation has an effect on Firm Value.

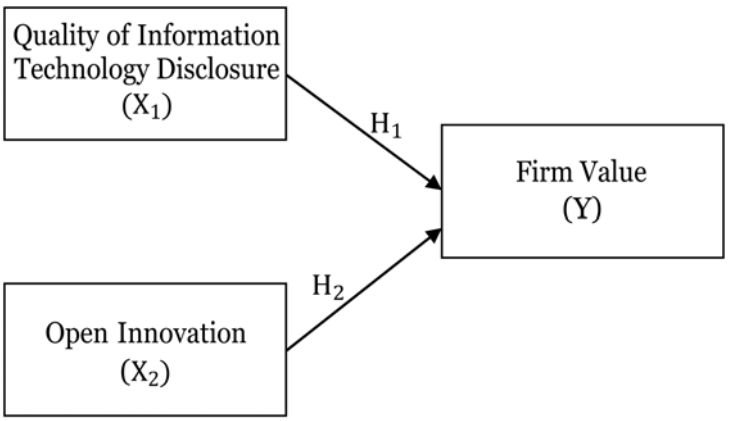


Figure 1. Research conceptual framework.

3 METHODOLOGY

This research is associative research by using the object of all go-public companies in the financial sector. The data is obtained from the annual report through the website www.idx.com and the official website of each company in the 2020–2022 period using documentation techniques. The sampling technique used purposive sampling with 3 criteria as follows:

(1) Companies in the financial sector that have been listed on the Indonesia Stock Exchange in 2020–2022
(2) Companies that publish annual reports
(3) Companies that have the required data in the study

Based on these criteria, 36 annual reports of financial sector companies were selected during the 2020–2022 period, thus obtaining 108 observation data.

3.1 *Operational definition of variables*

3.1.1 *Quality of information technology disclosure*

Information technology is defined as a branch of technology that is related to the knowledge, use and process of data in automatic collection, storing and organizing (Alalie *et al*. 2018).

This variable is measured by using the disclosure of information technology investment adapted from research conducted by (Kim and Lim 2011).

Table 1. Information technology investment disclosure index.

No	Items	Description
Quantitative Disclosure		
1.	IT Spending	It includes past, current or future IT spending, changes in IT spending, percentage of other performance measures and comparison with competitors' spending.
2.	IT Productivity	Included actual achievements of IT development, e.g. output, efficiency, and decrease in machine time or human resources.
3.	IT Returns	Considering actual IT management achievements beyond IT development, such as revenue, growth, profitability, market returns and other accounting ratios.
Qualitative Disclosure		
1.	IT Status Project	Through the proposed timing of the IT launch, estimated time of completion, progress or testing phase.
2.	IT Acquisition	Information technology is acquired by companies through purchase, adoption or development.
3.	Potential	Potential opportunities or risks from technology, strategy and organizational prospects.
4.	Market	The influence of market and environmental conditions that lead companies to invest in IT, for example through legal compliance, company- or market-specific needs, or economic trends.
5.	Market	Competition at the industry level. For example, the estimated market size for IT projects.
6.	Contract	Potential achievements from contractual relationships with customers or suppliers.

Source: Kim and Lim (2011).

The quality index of information technology investment disclosure is measured using an unweighted dichotomous scale with the following formula:

$$\textbf{IT ID} = \frac{\text{Number of disclosures made by the company}}{\text{Total number of disclosures that should be}} \times 100\%$$

3.1.2 *Open innovation*

Open Innovation is a process where companies can obtain ideas, technology, knowledge and skills from external parties such as academics, communities and other companies. This variable is measured using the formula adopted from research conducted by Berzkalne and Zelgalve (2013):

$$\textbf{Innovation} = \frac{\text{Intangible Assets}}{\text{Total Assets of the Company}}$$

3.1.3 *Firm value*

Firm value reflects investors' views of the company, measured through the stock price represented by Tobin's Q ratio. The measurement formula for the firm value variable is adopted from research conducted by Rinofah *et al.* (2023).

$$\textbf{Tobin's Q} = \frac{\text{MVE} + \text{D}}{\text{TA}}$$

Explanation:
Q = Tobin's Q
MVE = Market value (year-end closing stock price x number of shares outstanding at year-end)
D = Total debt (liabilities)
TA = Total assets

This study used a panel data regression analysis technique and was conducted using STATA version 15 test software for each test conducted. The data analysis process is carried out in the following steps:

(1) Tabulate the research data.
(2) Perform descriptive statistics.
(3) Determining the panel data regression model using three approaches, namely the Chow test, Hausman test and Lagrange Multiplier test.
(4) Conduct a classic assumption test. In this study, the classic assumption test used is the multicollinearity test, because the best model in this study is Random Effect. The Random Effect model uses the Generalized Least Square (GLS) approach, where the GLS approach does not need to use the classical assumption test because it is considered the Best Linear Unbiased Estimator (BLUE) (Napitupulu *et al.* 2021).
(5) Conduct a Goodness of Fit test of the regression model used.

4 RESULTS AND DISCUSSION

4.1 *Descriptive statistical test results*

Table 2. Descriptive statistical test results.

Variable	Obs.	Mean	Median	Std. Dev	Min	Max
Y NP	108	1.533192	1.007103	2.086276	0.7346515	18.49779
X1 TI	108	0.404321	0.4444444	0.1667243	0.1111111	0.7777778
X2 OI	108	0.0040451	0.0028331	0.0041846	0.0000553	0.0273227

Source. Data processed, 2023.

The results for the first independent variable, the Quality of Information Technology Disclosure, have a mean of 0.404321 and a standard deviation of 0.1667243, which means that as many as 40% of companies have disclosed information about information technology.

The second independent variable, Open Innovation, has a mean of 0.0040451 or 0.4% with a standard deviation of 0.0041846.

The dependent variable, Firm Value, obtained the highest mean result of 1,533192 with a standard deviation of 2,086276.

4.2 *Determination of panel data regression model Chow test*

Table 3. Chow test results.

Effect Test	Probability
Prob > F	**0.0007**

Source. Data processed, 2023.

If the resulting value in Prob > F > the significance level, it means that the suitable model to use is the Common Effect Model, and vice versa. Based on the table above, it can be seen that the Prob> F value = 0.0007 < 0.05, meaning that the best choice is the Fixed Effect Model so it is continued with the Hausman test.

4.3 *Hausman test*

Table 4. Hausman test results.

Effect Test	Probability
Prob>chi2	**0.0859**

Source. Data processed, 2023.

Based on the data results shown in Table 4. by looking at Prob> chi2 = 0.0859 > 0.05 which means that the probability is bigger than the 5% significance level so the best model is used by the Random Effect Model.

4.4 *Lagrange multiplier test*

Table 5. Lagrange multiplier test results.

Effect Test	Probability
Prob > chibar2	**0.0018**

Source. Data processed, 2023.

Based on the data results displayed in Table 5. by looking at Prob> chibar2 = 0.0018 > 0.05 which means that the probability is smaller than the 5% significance level so the best model is used by the Random Effect Model.

4.5 *Classical assumption test multicollinearity test*

The multicollinearity test aims to test whether the regression model found a correlation between independent variables (independent). The way to detect multicollinearity in the regression model is to look at the strength of the correlation between the independent variables, if there is a correlation between the independent variables> 0.90, it can indicate the presence of multicollinearity.

Table 6. Multicollinearity test results.

correlate TI OI (obs = 108) TI	OI TI	OI
TI	1.0000	
OI	0.1329	1.0000

Source. Data processed, 2023.

Based on the table above, it can be seen that the correlation between X1 (TI) and X2 (OI) is 0.1329 < 0.90, which means that the regression model does not have multicollinearity.

4.6 *Goodness of fit panel data regression model*

Based on the test on determining the panel data regression model through 3 tests, it is known that the suitable model used in this study is the Random Effect Model.

Table 7. Random effect regression test model results.

Variabel	Coefficient	T-stat	Prob.	Keputusan
X1 TI	−1.58518	−1.30	0.195	H_1 Rejected
X2 OI	135.3901	2.52	0.012	H_2 Accepted
_cons	1.626448	2.88	0.004	
		Weighted Statistics		
Adj. R Squared			0.1301	
F-statistic			7.18	
Prob. (F-statistic)			0.0275	

Source. Data processed, 2023.

4.6 *Coefficient of determination (R^2)*

The coefficient of determination (R^2) is used to measure how far the model's ability to explain variations in the dependent variable. Based on the data in Table 7, it is stated that the value of the coefficient of determination is 0.1301, which means that there is a relationship between the independent variable and the dependent variable. The value of Adj. R-Square value is 0.1301, which means that all the independent variables used in this study can explain the dependent variable by 0.1301 or 13%.

4.7 *Hypothesis test simultaneous test (F test)*

Based on the data shown in Table 7, it is known that the probability value of the F-statistic is 0.0275 <0.05. With the acceptance of H_1, it can be interpreted that the Quality of Information Technology Disclosure and Open Innovation simultaneously affect the Firm's Value.

4.8 *Partial test (t Test)*

The test is conducted on the population regression coefficient, whether it is equal to zero, which means that the independent variable has no significant effect on the dependent variable. The explanation for Table 7 is below:

4.8.1 *Quality of information technology disclosure*

The first hypothesis shows that the variable quality of information technology disclosure does not affect firm value. The probability value is 0.195 and the value is bigger than 0.05. It can be concluded that the independent variable, the quality of information technology disclosure with 108 samples of financial sector companies used, does not affect firm value.

4.8.2 *Open innovation*

The second hypothesis shows that the Open Innovation variable affects firm value. The probability value is $0.012 < 0.05$. It can be concluded, the independent variable, Open Innovation with 108 samples of financial sector companies used influences firm value.

4.9 *The effect of information technology disclosure quality on firm value*

Financial sector companies tend to be the first to adopt information technology compared to other sectors because the financial industry is highly dependent on information (Achjari and

Wahyuningtyas 2014). Therefore, financial sector companies tend to invest and announce information about IT more frequently than non-financial companies. However, the signals provided by companies through annual reports about information technology investment are not considered by investors so the research results on this variable have no impact on firm value. This is supported by a statement from Achjari and Wahyuningtyas (2014) that, unlike foreign investors, investors in Indonesia do not see IT investment announcements as a basis for investment decisions. So, it can be concluded that investors do not consider the information provided by the company in developing digital. Therefore, information about IT investment has not been able to convince investors so it does not have an impact on firm value.

Although PT Bank Mandiri Tbk has disclosed higher IT investments totaling 5 to 7 items from a total of 9 items, it is not followed by an increase in firm value. In contrast, PT Allo Bank Indonesia Tbk which tends to disclose 1 to 4 items for 3 years has a high firm value. Therefore, whether many or few disclose information about IT investment, it does not necessarily have an impact on increasing the company value of an issuer. The results of this study are supported by research conducted by Andrayani and Dewi (2014).

4.10 *The effect of open innovation on firm value*

Investors tend to be interested in companies that have a competitive advantage to maintain their position and gain profit from the investment they make. (Aditya and Kaswar 2022). In addition, investors also tend to be interested in companies that manage their invested funds well, especially if the funds are allocated for innovation activities (Nohong 2016).

Where if the value of the intangible asset ratio is 1% or more than 1%, innovation can increase the value of the firm (Berzkalne and Zelgalve 2013). 37 samples have intangible asset ratio values of 1% or more than 1%. Research results from Emeneka and Okerekeoti (2022) and Berzkalne and Zelgalve (2013) also found that innovation in intangible assets has a significant effect on firm value. In addition, the same results were found by Choi *et al.* (2022) which shows that Open Innovation has a significant effect on firm value.

5 CONCLUSION

Based on the results of the goal formulation and hypothesis testing in this study, it can be concluded that the quality of information technology disclosure is unable to influence firm value. because investors do not consider disclosure of information related to information technology in the future so they rarely see important factors such as investments made by companies as long-term efforts to increase their firm value in the future.

Open Innovation has a significant effect on firm value. Investors get positive signals from the company regarding collaboration with partners in developing new product or service innovations, which attracts them to invest and 37 samples produce a value of 1% or more than 1%. This can be interpreted that open innovation, which is collaborating with external partners by the company, increases the firm's value.

This study has limitations, which is the measurement of the variable quality of information technology disclosure using an unweighted dichotomous scale (1 and 0) for each disclosure item even though the way of disclosing the information is not the same so that the results obtained have no effect due to the lack of strength of the measurements used. So that further research can use the same measurement but use a different scale, that is with a rating scale, such as giving a value of 1 to briefly disclose one item, a value of 2 to disclose and provide a qualitative explanation, a value of 3 if the item is disclosed, given a qualitative explanation and mentions a nominal number and so on.

This study focuses on measuring Open Innovation using intangible assets because, in the financial sector, not all companies disclose R&D costs which are often used as a proxy for

measuring firm innovation. However, many companies do not disclose their intangible assets, which also reduces the population in this study. So that further research can use other measurements such as using R&D costs, dummy variables for market partnerships and science partnerships and so on to get stronger results to prove the effect of Open Innovation on firm value.

BIBLIOGRAPHY

Alalie, H. M., Harada, Y., & Noor, I. M. (2018). A resource-based view: How information technology creates sustainable competitive advantage to improve organizations. *Journal of Advance Management Research*, *6* (12), 1–5.

Andrews, D., & De Serres, A. (2012). *Intangible Assets, Resource Allocation and Growth: A Framework for Analysis*. Economics Department Working Papers No. 989. https://doi.org/10.1787/18151973

Annisa Indah Mutiasari. (2020). Perkembangan industri perbankan di era digital. *Ekonomi Bisnis dan Kewirausahaan*, *9*(2), 32–41. https://doi.org/10.47942/iab.v9i2.541

Arya Aji Aditya, & Andi Rahmat Kaswar. (2022). The relationship between innovation capital, firm value, and firm risk. *EkBis: Jurnal Ekonomi Dan Bisnis*, *6*(1), 74–86. https://doi.org/10.14421/EkBis.2022.6.1.1558

Bardhan, I., Krishnan, V., & Lin, S. (2013). Business value of information technology: Testing the interaction effect of IT and R&D on Tobin's Q. *Information Systems Research*, *24*(4), 1147–1161. https://doi.org/10.1287/isre.2013.0481

Berzkalne, I., & Zelgalve, E. (2013). Innovation and company value: Evidence from the Baltic countries. *Regional Formation and Development Studies*, *3*(11), 39–51. https://doi.org/10.15181/rfds.v11i3.608

Catalin Ionita, & Elena Dinu. (2021). The effect of intangible assets on sustainable growth and firm value – Evidence on intellectual capital investment in companies listed on Bucharest Stock Exchange. *Kybernetes*, *50*(10), 2823–2849. https://doi.org/10.1108/k-05-2020-0325

Chesbrough, H. W. (2003). *Open Innovation: The New Imperative for Creating and Profiting from Technology*. Boston: Harvard Business Press.

Choi, Y. K., Lim, J., & Jeon, S. (2022). The impact of a platform company's open innovation activities on its firm value: In the perspective of the venture ecosystem. *Technology Analysis and Strategic Management*, *34* (3), 322–334. https://doi.org/10.1080/09537325.2021.1896699

Didi Achjari, & Annisa Eka Wahyuningtyas. (2014). Information technology investment announcements and firms' value: The case of Indonesian firms in the financial and non-financial sectors. *Asian Journal of Business and Accounting*, *7*(2), 95–116.

Eka Putri Primawanti, & Hapzi Ali. (2021). Pengaruh teknologi informasi, sistem informasi berbasis web dan knowledge management terhadap kinerja karyawan (literature review executive support sistem (ess) for business). *JEMSI: Jurnal Ekonomi Manajemen Sistem Informasi*, *3*(3), 267–285. https://doi.org/10.31933/jemsi.v3i3.818

Emeneka, O. L., & Okerekeoti, C. U. (2022). The intangible assets innovation and firm value of quoted consumer goods manufacturing firms in nigeria. *Journal of Accounting and Financial Management*, *8*(7), 2695–2211. https://doi.org/10.56201/jafm.v8.no7.2022.pg225.234

Erica Mazzola, Manfredi Bruccoleri, & Giovanni Perrone. (2016). Open innovation and firms' performance: state of the art and empirical evidences from the bio-pharmaceutical industry. *Int. J. Technology Management*, *70*(2/3), 109–134. https://doi.org/10.1504/IJTM.2016.075152

Farouk, B. K. U., & Dandago, K. I. (1970). Impact of investment in information technology on financial performance of nigerian banks: Is there a productivity paradox? *The Journal of Internet Banking and Commerce*, *20*(1), 1–22.

Grimaldi, M., Corvello, V., De Mauro, A., & Scarmozzino, E. (2017). A systematic literature review on intangible assets and open innovation. *Knowledge Management Research & Practice*, *15*(1), 90–100. https://doi.org/10.1057/s41275-016-0041-7

Irene Andrayani, & Anastasia Fenyta Dewi. (2014). Pengaruh it spending terhadap kinerja perusahaan pada perusahaan telekomunikasi yang terdaftar di bursa efek seluruh asia tenggara pada tahun 2009–2011. *Modus*, *26*(2), 93–106. https://dx.doi.org/10.24002/modus.v26i2.581

Karim, S., Qamruzzaman, M., & Jahan, I. (2022). Nexus between information technology, voluntary disclosure, and sustainable performance: What is the role of open innovation? *Journal of Business Research*, *145*, 1–15. https://doi.org/10.5281/zenodo.7540292

Kim, J. W., & Lim, J. H. (2011). IT investment disclosure, information quality, and factors influencing managers' choices. *Information & Management*, *48*(2–3), 114–123. https://doi.org/10.1016/j.im.2011.03.001

Lars Witell, Heiko Gebauer, Elina Jaakkola, Wafa Hammedi, Lia Patricio, & Helen Perks. (2017). A bricolage perspective on service innovation. *Journal of Business Research*, *79*, 290–298. https://doi.org/10.1016/j.jbusres.2017.03.021

Maharani, S. A., & Puspitasari, E. (2022). Pengaruh profitabilitas, leverage dan biaya penelitian dan pengembangan terhadap nilai perusahaan. *Jurnal Akuntansi Dan Pajak*, *22*(2), 962–970. http://dx.doi.org/10.29040/jap.v22i2.2860

Masculine Muhammad Muqorobin, Ayu Anggraini, Ayu Dwi Rahmawati, Diana Yohanes, & Faricha Dzakia Ifkarina. (2021). Pengaruh open banking berbasis open API terhadap eksistensi perbankan. *MAKSIMUM: Media Akuntansi Universitas Muhammadiyah Semarang*, *11*(2), 75–84. https://doi.org/10.26714/mki.11.2.2021.75-84

Maulandy Rizki Bayu Kencana. *Kinerja Saham Bank Digital Amblas di 2022, Ini Penyebabnya. Liputan6. Com*. https://www.liputan6.com/ [diakses pada 18 Juli 2023]

Mursalim Nohong. (2016). Inovasi, pertumbuhan, ukuran dan nilai Perusahaan farmasi di Indonesia. *Jurnal Keuangan Dan Perbankan*, *20*(2), 176–185. https://doi.org/10.26905/jkdp.v20i2.250

Nanda Dwi Fitri, Suherman, & Sholatia Dalimunthe. (2023). Does innovation affect firm value study on compas100 index companies for the 2016 – 2020 Period. *International Journal of Education, Social Studies, And Management (IJESSM)*, *3*(1), 80– 88. https://doi.org/10.52121/ijessm.v3i1.139

Pertiwi, S. P., & Suhartini, D. (2022). pengaruh diversifikasi pendapatan, modal intelektual dan pengungkapan manajemen risiko terhadap nilai perusahaan (Studi empiris pada bank umum konvensional yang terdaftar di bursa efek Indonesia tahun 2015–2020). *Fair Value: Jurnal Ilmiah Akuntansi Dan Keuangan*, *5* (2), 2622–2205. https://doi.org/10.32670/fairvalue.v4i7.1179

Risa, S. A., & Hermanto, S. B. (2023). Pengaruh teknologi informasi dan corporate governance terhadap nilai perusahaaN. *Jurnal Ilmu Dan Riset Akuntansi*, *12*(1), 1–19.

Risal Rinofah, Ratih Kusumawardani, & Prahadhita Nerissa Putri. (2023). pengaruh corporate social responsibility terhadap nilai perusahaan dengan profitabilitas sebagai variabel moderating (studi empiris pada perusahaan manufaktur yang terdaftar di bursa efek indonesia periode 2015–2019). *Jurnal Manajemen Terapan Dan Keuangan (Mankeu)*, *12*(1), 40–52. https://doi.org/10.22437/jmk.v12i01.17086

Runggu Besmandala Napitupulu, Torang P. Simanjuntak, Lamminar Hutabarat, Hormaingat Damanik, Hotriado Harianja, Ronnie Togar Mulia Sirait, & Chainar Elli Ria Lumban Tobing. (2021). *Penelitian Bisnis Teknik dan Analisis Data dengan SPSS, STATA, EVIEWS* (1st ed.). Medan: Madanatera.

Sengupta, A., & Sena, V. (2020). Impact of open innovation on industries and firms – A dynamic complex systems view. *Technological Forecasting and Social Change*, 1–35. https://doi.org/10.1016/j.techfore.2020.120199

Spence, M. (1973). Job market signaling. *The Quarterly Journal of Economics*, *87*(3), 355–374. https://doi.org/10.2307/1882010

Ulum, I., Juanda, A., & Leniwati, D. (2021). *Metodologi Penelitian Akuntansi* (Edisi 3). Malang: Penerbit Baskara Media Aditya Media Group.